

```

#include<iostream>
//Take a number from the user. Print each digit separately.
//Output is individual digits printed.
//Process % to extract the last digit and / to remove the last digit.
//Input is: read the number from the user.
using namespace std;
int main()
{
    int number,digit;
    cout<<"Enter a number:";
    cin>>number;
    while(number>0)
    {
        digit=number%10;
        number=number/10;
        cout<<"\nNumber left:"<<number<<"\tDigit:"<<digit;
    }
    return 0;
}

```

```

#include<iostream>
//Write a program Check whether a given number is Armstrong or not.
//Given a three-digit number, an Armstrong number is a number where,
//if the individual digits are cubed and added together,
//the number itself will be regenerated.
//153--->1^3+5^3+3^3-->153
//If sum = number then it is Armstrong.
//We need to maintain a number.
//Accept a number--->create a copy of it,
//calculate the sum, and then compare with the original.
using namespace std;
int main()
{
    int number,digit,sum=0;
    cout<<"Enter a number:";
    cin>>number;
    while(number>0)
    {
        digit=number%10;
        number=number/10;
        sum=sum+digit;
    }
    cout<<"Sum of digit is : "<<sum;
    return 0;
}

```

```

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```

```

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//Accept a number--->create a copy of it,
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using namespace std;
int main()
{

```

```

    int number,r_number=0,digit;
    cout<<"Enter a number:";
    cin>>number;
    while(number>0)//to find sum
    {
        digit=number%10;
        r_number=r_number*10+digit;
        number=number/10;
    }
    cout<<"Reverse number is : "<<r_number;

    return 0;
}

```

```

#include<iostream>

```

```

using namespace std;
int main()
{
    int number,min,max,digit;
    //Read a number from the user.
    cout<<"\nEnter a number:";
    cin>>number;//592617
    //Assign last digit as min and max.
    min=max=number%10;
    //Use a while loop till the number is not zero. Remove one digit at a time.
    //Compare with min and max.
    //If less than minimum, update minimum. If more than maximum, update maximum.
    while(number>0)
    {
        digit=number%10;
        if(digit>max)
        {
            max=digit;
        }
        else if(digit<min)
        {

```

```

        min=digit;
    }
    number=number/10;
}
cout<<"\nFrom a given number, min:"<<min<<" and max:"<<max;
return 0;
}

```

ATM PIN Digit Analyzer

A security system receives a numeric code. Write a program using a while loop to determine:

Number of digits

Sum of digits

Largest digit

Smallest digit

Number of even digits

Example:

Input: 583927

Output:

Digits = 6

Sum = 34

Largest = 9

Smallest = 2

Even Digits = 2

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        //menu
```

```
        cout<<"\nApni Taprii ";
```

```
        cout<<"\n1.Tea";
```

```
        cout<<"\n2.Coffee";
```

```
        cout<<"\n3.Water";
```

```
        cout<<"\n0.Exit";
```

```
        cout<<"\n:";
```

```
        //input
```

```
        cin>>choice;
```

```
        //switch to do taks
```

```
        switch(choice)
```

```
        {
```

```
            case 1:
```

```

        cout<<"\nYou selected Tea";
        break;
    case 2:
        cout<<"\nYou selected Coffee";
        break;
    case 3:
        cout<<"\nYou selected Water";
        break;
    case 0:
        cout<<"\nExiting system. Thanks for using it. ";
        break;
    default:
        cout<<"\nInvalid choice.";
        break;
    }
}while(choice!=0); //Till exit option is not selected.
return 0;
}

```

Modify the above program with a slight change. Now when a user selects any item (tea, coffee, or water), a question should pop up for quantity and then a bill should be generated, considering:

- tea is ₹20
- coffee is ₹60
- water is ₹15

Example: if a user selects tea, ask the question, "How many teas?" If the user specifies 5 × 20, that is ₹100 Display message: amount payable is ₹100.

```
#include<iostream>
```

```

using namespace std;
int main()
{
    int choice;
    do
    {
        //menu
        float cost_tea=20.00,cost_coffee=60.00,cost_water=15.00;
        int qty;
        cout<<"\nApni Taprii ";
        cout<<"\n1.Tea";
        cout<<"\n2.Coffee";
        cout<<"\n3.Water";
        cout<<"\n0.Exit";
        cout<<"\n:";
        //input
        cin>>choice;
    }
}

```

```

//switch to do tasks
switch(choice)
{
    case 1:
        cout<<"\nEnter number of teas:";
        cin>>qty;
        cout<<"\nYou order total "<<qty<<" teas cost:"<<qty*cost_tea;
        break;
    case 2:
        cout<<"\nEnter number of Coffees:";
        cin>>qty;
        cout<<"\nYou order total "<<qty<<" coffee
cost:"<<qty*cost_coffee;
        break;
    case 3:
        cout<<"\nEnter number of Water Bottles:";
        cin>>qty;
        cout<<"\nYou order total "<<qty<<" Water bottles
:"<<qty*cost_water;
        break;
    case 0:
        cout<<"\nExiting system. Thanks for using it. ";
        break;
    default:
        cout<<"\nInvalid choice.";
        break;
}
}while(choice!=0);//Till exit option is not selected.
return 0;
}

```

Write a program to simulate an entire calculator capable of doing addition, subtraction, multiplication, and division.

Normal flow user can use:

1. Option 1 to set two numbers
2. Option 2 to add them
3. Option 3 to subtract
4. Option 4 to multiply
5. Option 5 to divide
6. Option 0 to exit

By default users will first set to numbers. If not then by default both are zero. The result should be shown as a mathematical calculation.

1----->enter 2 numbers:

10 20

Number set

2----->addition-----> 10+20=30

```

#include<iostream>

using namespace std;
int main()
{
    int choice;
    float no1=0,no2=0;
    do
    {
        //menu
        cout<<"\nMy Caculator";
        cout<<"\n1.Set numbers";
        cout<<"\n2.Add";
        cout<<"\n3.Sub";
        cout<<"\n4.Mul";
        cout<<"\n5.Div";
        cout<<"\n0.Exit";
        cout<<"\n:";
        //input
        cin>>choice;
        //switch to do taks
        switch(choice)
        {
            case 1:
                cout<<"\nEnter 2 numbers ";
                cin>>no1>>no2;
                cout<<"\nno1:"<<no1<<" no2:"<<no2;
                break;

            case 2:
                cout<<"\n"<<no1<<" + "<<no2<<" = "<<(no1+no2);
                break;

            case 3:
                cout<<"\n"<<no1<<" - "<<no2<<" = "<<(no1-no2);
                break;

            case 0:
                cout<<"\nExiting system. Thanks for using it. ";
                break;

            default:
                cout<<"\nInvalid choice.";
                break;
        }
    }while(choice!=0);//Till exit option is not selected.
    return 0;
}

```

```

#include<iostream>
//1,2,3...,10
using namespace std;
int main()
{
    for (int i=1;i<=10;i++)
    {
        cout<<i<<" ";
    }
    return 0;
}

```

//Write a program to accept a number from the user and print a multiplication table from 1 to 10.

Input:2

no i no*i

2 X 1=2

2 X 2=4

.

.

.

.

2 X 10=20

```

#include<iostream>
//1,2,3...,10
using namespace std;

int main()
{
    int n,sum=0;
    cout<<"Enter number:";
    cin>>n;
    for (int i=1;i<=n;i++)
    {
        cout<<i<<" ";
        sum=sum+i;
    }
    cout<<"="<<sum;
    return 0;
}

```

Modify the above program similarly to print the factorial of a number or the product up to n of a number.

no=4 1X2X3X4X=24

```

#include<iostream>
//Write a program Check whether a given number is Prime or not.
using namespace std;
int main()
{
    int no;
    cout<<"Enter number to check:";
    cin>>no;
    bool flag=true;//Think positively that it is a prime number.
    for(int i=2;i<no;i++)
    {
        cout<<"\nchecking with "<<no<<"%"<<i;
        if(no%i==0)
        {
            flag=false;
            break;
        }
    }
    if(flag==true)
        cout<<endl<<no<<"is prime";
    else
        cout<<endl<<no<<" is not prime";
    return 0;
}

```

Modify the above program to print all the prime numbers from the range of 1 to 50.

```

#include<iostream>
//Write a program which will continue accepting numbers
//till the user gives 0 / a negative number to stop.
using namespace std;
int main()
{
    int number;
    while(true)
    {
        cout<<"Enter number or 0 to stop:";
        cin>>number;
        if(number<=0)
        {
            cout<<"\nEnding the system. ";
            break;
        }
    }
    cout<<"outside loop :bye bye";
    return 0;
}

```

Modify the above program to accept numbers till the user hits 0 and keep on counting whether the entered number by the user is even or odd.